11. Sequences and Series

11.10. Taylor and Maclaurin Series

11.10 Learning Objectives
- Find the Taylor Series Expansion of a function
- Find the Maclaurin Series Expansion of a function
- Find the function represented by the Taylor or Maclaurin Series
- Use known Maclaurin Series Expansion of functions to describe complicated functions

11.10.1. Maclaurin Series of Common Functions

You should be able to recognize (or re-derive) the following functions and their series. The first 3 are derived from 11.8 techniques (using geometric series and/or $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$). The others come from $c_n = \frac{f^{(n)}(a)}{n!}$.

Function	Series	First Few Terms	Interval of Convergence
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$1 + x + x^2 + x^3 + x^4 + \ldots$	$(-1, 1)$
$\arctan(x)$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots$	$(-1, 1)$
$\ln(1+x)$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$	$(-1, 1)$
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$	$(-\infty, \infty)$
$\cos(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$	$(-\infty, \infty)$
$\sin(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$	$(-\infty, \infty)$

Exercise 11.10.1

Derive the above Maclaurin series expansions and intervals of convergence for the above series. You will need ratio test for some of these.

Exercise 11.10.2

Some of the functions above are "odd" ($f(-x) = -f(x)$) or "even" ($f(x) = f(-x)$) functions. Which are odd and even. What do you notice about the powers on the Maclaurin series expansions of these functions?

Exercise 11.10.3

Sketch (by hand or desmos) a graph of $y = \cos(x)$ and the second order Taylor Polynomial (centered at 0) of $\cos(x)$ ($T_2(x) = 1 - x^2$) (Sum the series only until you have a degree 2 polynomial, then ignore the other terms) on the same graph. What do you observe?

Sketch a graph of $y = \sin(x)$ and the first order Taylor Polynomial (centered at 0) of $\sin(x)$. What do you observe?

How can your observations help you remember the Maclaurin series expansions for $\sin(x)$ and $\cos(x)$?

Example 11.10.1

Use the Maclaurin series of $\sin(x)$ to evaluate $\displaystyle\int \frac{\sin(x)}{x}\,\mathrm{d}x$.

Note

Notice that we can't compute this integral manually using techniques from this class. Not even a double integration by parts step will let us evaluate this! Switching this to a series problem lets us actually compute the integral however!

Solution

Since

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

we have

$$\begin{aligned}
\int \frac{\sin(x)}{x}\,\mathrm{d}x &= \int \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}\,\mathrm{d}x \\
&= \int \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1-1}}{(2n+1)!}\,\mathrm{d}x \\
&= \sum_{n=0}^{\infty} \frac{(-1)^n \int x^{2n}\,\mathrm{d}x}{(2n+1)!} \\
&= C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}
\end{aligned}$$

Example 11.10.2

Using the Maclaurin series of $\sin(x)$, evaluate the following limit

$$\lim_{x \to 0} \frac{2\sin(x^2) - 2x^2 + \frac{1}{3}x^6}{x^{10}}$$

Warning

Since $x \to 0$ instead of $x \to \infty$, we cannot look at the dominating terms for this limit. Turning to series lets us evaluate this limit!

Solution

Notice

$$\begin{aligned}\sin(x^2) &= \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!} \\ &= x^2 - \frac{x^6}{6} + \frac{x^{10}}{120} + \sum_{n=3}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}\end{aligned}$$

so

$$\begin{aligned}\lim_{x\to 0} \frac{2\sin(x^2) - 2x^2 + \frac{1}{3}x^6}{x^{10}} &= \lim_{x\to 0} \frac{2\left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}\right) - 2x^2 + \frac{1}{3}x^6}{x^{10}} \\ &= \lim_{x\to 0} \frac{2\left(x^2 - \frac{x^6}{6} + \frac{x^{10}}{120} + \sum_{n=3}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}\right) - 2x^2 + \frac{1}{3}x^6}{x^{10}} \\ &= \lim_{x\to 0} \frac{2x^2 - \frac{x^6}{3} + \frac{x^{10}}{60} + \sum_{n=3}^{\infty} \frac{(-1)^n 2x^{4n+2}}{(2n+1)!} - 2x^2 + \frac{1}{3}x^6}{x^{10}} \\ &= \lim_{x\to 0} \frac{\frac{x^{10}}{60} + \sum_{n=3}^{\infty} \frac{(-1)^n 2x^{4n+2}}{(2n+1)!}}{x^{10}} \\ &= \lim_{x\to 0} \frac{1}{60} + \sum_{n=3}^{\infty} \frac{(-1)^n 2x^{4n+2-10}}{(2n+1)!} \\ &= \frac{1}{60}\end{aligned}$$

Example 11.10.3

Find the sum of the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1}}{3^{2n+1}(2n+1)!}$$

Solution

This is not a geometric series, so our only other way to evaluate this series is to match it up with a series we know. Because $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$ we can see

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1}}{3^{2n+1}(2n+1)!} &= \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{\pi}{3}\right)^{2n+1}}{(2n+1)!} \\ &= \sin\left(\frac{\pi}{3}\right) \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

11.10a Section Summary:

- We went over common Maclaurin series that you can identify. (Make flashcards for these series now and/or be able to re-derive them quickly.)
- We used the Taylor series/Maclaurin series of functions to compute integrals and limits that we couldn't evaluate otherwise.